

Notes on Asymptotic Analysis

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1 Information Inequality

Theorem 1. Let $\eta_n := \eta_n(X_1, \dots, X_n)$ be estimator for $\tau := \tau(\theta)$. Then, under R0-R5,

$$\text{Var}_\theta(\eta_n) \geq \nabla \mathbb{E}_\theta(\eta_n)^\top \{nI(\theta)\}^{-1} \nabla \mathbb{E}_\theta(\eta_n)$$

Proof. Note that we have

$$\begin{aligned} \text{Cov}_\theta(\eta_n, S_n(\theta)) &= \mathbb{E}_\theta[\eta_n S_n(\theta)^\top] \\ &= \int \eta_n (\nabla \ell_n(\theta))^\top \prod_{i \in [n]} f_\theta(x_i) dx_1 \dots dx_n \\ &= \int \eta_n (\nabla \ell_n(\theta))^\top \exp(\ell_n(\theta)) dx_1 \dots dx_n \\ &= \int \eta_n \{\nabla \exp(\ell_n(\theta))\}^\top dx_1 \dots dx_n \\ &= \nabla \left\{ \int \eta_n \exp(\ell_n(\theta)) dx_1 \dots dx_n \right\} \\ &= \nabla \mathbb{E}_\theta[\eta_n]^\top \end{aligned}$$

The best linear predictor of η_n in terms of $S_n(\theta)$ is

$$\begin{aligned} &\mathbb{E}_\theta(\eta_n) + \text{Cov}_\theta(\eta_n, S(\theta)) \{ \text{Var}_\theta[S(\theta)] \}^{-1} \{ S_n(\theta) - \mathbb{E}_\theta[S_n(\theta)] \} \\ &= \mathbb{E}_\theta(\eta_n) + \nabla \mathbb{E}_\theta[\eta_n]^\top \{nI(\theta)\}^{-1} S_n(\theta) \\ &= a + B S_n(\theta) \end{aligned}$$

The variance of the difference of η_n and its best linear predictor is then

$$\begin{aligned} &\text{Var}_\theta[\eta_n - \mathbb{E}_\theta(\eta_n) - \nabla \mathbb{E}_\theta[\eta_n]^\top \{nI(\theta)\}^{-1} S_n(\theta)] \\ &= \text{Var}_\theta(\eta_n) + B \text{Var}_\theta(S_n(\theta)) B^\top - \text{Cov}_\theta(\eta_n, S_n(\theta)) B^\top - B \text{Cov}_\theta(S_n(\theta), \eta_n) \\ &= \text{Var}_\theta(\eta_n) - \nabla \mathbb{E}_\theta[\eta_n]^\top \{nI(\theta)\}^{-1} \nabla \mathbb{E}_\theta[\eta_n] \geq 0 \end{aligned}$$

□

Corollary 1. If η_n is an unbiased estimator for $\tau := \tau(\theta)$,

$$\text{Var}_\theta(\eta_n) \geq \nabla \tau(\theta)^\top \{nI(\theta)\}^{-1} \nabla \tau(\theta)$$

Proof. It is straightforward by the fact that $\mathbb{E}_\theta[\eta_n] = \tau(\theta)$

□

Corollary 2 (Cramer-Rao Inequality). *If $\hat{\theta}_n$ is an unbiased estimator for θ ,*

$$\text{Var}_{\theta}(\hat{\theta}_n) \succeq \{nI(\theta)\}^{-1}$$

Proof. Say $\eta := c^{\top} \theta$ for any c , and consider $\eta_n = c^{\top} \hat{\theta}_n$. Then, by the Theorem 1, we have

$$\text{Var}_{\theta}(\eta_n) = c^{\top} \text{Var}_{\theta}(\hat{\theta}_n) c \geq c^{\top} \{nI(\theta)\}^{-1} c,$$

which gives us the desired result. $\square$

2 Big/Little-Oh in a Probability Language

Let $\{X_n\}$ and $\{R_n\}$ be sequences of random variables, and $\{a_n\}$ be a corresponding sequences of constants.

Definition 1 (Big-O-p Notations). We say X_n/R_n is bounded in probability and denoted $X_n := O_p(R_n)$ if

$$\forall \varepsilon > 0, \exists M < \infty, \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N, \mathbb{P}(|X_n/R_n| > M) < \varepsilon.$$

Below statement is equivalent.

$$\lim_{k \rightarrow \infty} \sup_n \mathbb{P}(|X_n/R_n| > k) = 0$$

Definition 2 (Little-O-p Notations). We denote $X_n := o_p(R_n)$ if

$$\left| \frac{X_n}{R_n} \right| \xrightarrow{P} 0$$

It suffices to work with the orders $O_p(1)$ and $o_p(1)$. Indeed, to compare two sequences of random variables (X_n) and (R_n) , one can reduce the problem to the ratio X_n/R_n (whenever $R_n \neq 0$ with high probability) and then study whether this ratio is $O_p(1)$ or $o_p(1)$.

By definition, $X_n = o_p(R_n) \iff X_n/R_n = o_p(1)$ and $X_n = O_p(R_n) \iff X_n/R_n = O_p(1)$.

Theorem 2. *Suppose cdf of X_n and X are continous function. Then,*

$$X_n \xrightarrow{d} X \Rightarrow X_n = O_p(1)$$

Proof. Suppose $X_n \xrightarrow{d} Z$. Let $\varepsilon > 0$ be given. Since Z is a random variable (property of cdf),

$$\lim_{k \rightarrow \infty} \mathbb{P}(|Z| > k) = 0.$$

Thus, there exists $k_0 > 0$ such that

$$\mathbb{P}(|Z| > k_0) < \frac{\varepsilon}{2}.$$

Since $X_n \xrightarrow{d} Z$, we have

$$\mathbb{P}(|X_n| > k_0) \rightarrow \mathbb{P}(|Z| > k_0).$$

Then, there exists $N \in \mathbb{N}$ such that for all $n > N$,

$$\begin{aligned} |\mathbb{P}(|X_n| > k_0) - \mathbb{P}(|Z| > k_0)| &< \frac{\varepsilon}{2} \Rightarrow \mathbb{P}(|X_n| > k_0) \leq \mathbb{P}(|Z| > k_0) + \frac{\varepsilon}{2} \\ \therefore \sup_{n \geq N} \mathbb{P}(|X_n| > k_0) &\leq \mathbb{P}(|Z| > k_0) + \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

Now fix the finitely many random variables $X_1, \dots, X_N$. For each $1 \leq n \leq N$,

$$\lim_{k \rightarrow \infty} \mathbb{P}(|X_n| > k) = 0,$$

so there exists $k_n > 0$ such that

$$\mathbb{P}(|X_n| > k_n) < \varepsilon.$$

Set

$$K := \max\{k_0, k_1, \dots, k_N\}.$$

Then for $1 \leq n \leq N$,

$$\mathbb{P}(|X_n| > K) \leq \mathbb{P}(|X_n| > k_n) < \varepsilon,$$

and for $n > N$,

$$\mathbb{P}(|X_n| > K) \leq \mathbb{P}(|X_n| > k_0) < \varepsilon.$$

Hence,

$$\sup_{n \geq 1} \mathbb{P}(|X_n| > K) \leq \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, it follows that X_n is bounded in probability. □

Examples

- If $\sqrt{n}(X_n - \theta) \xrightarrow{d} N(0, \sigma^2)$, $X_n = \theta + O_p(n^{-1/2})$.

Theorem 3. Let X_n, Y_n and Z_n be random variables such that

$$X_n = O_p(1), \quad Y_n = O_p(1), \quad Z_n = o_p(1).$$

Then, as $n \rightarrow \infty$,

1. $X_n + Y_n = O_p(1)$,
2. $X_n Y_n = O_p(1)$,
3. $Z_n X_n = o_p(1)$,
4. $Z_n = O_p(1)$,
5. $Z_n + X_n = O_p(1)$. (This follows immediately from 4.)

Theorem 4. If $f(x) = o(g(x))$ as $x \rightarrow a$ and $X_n \xrightarrow{P} a$, then $f(X_n) = o_p(g(X_n))$

Proof. Let $\epsilon_1, \epsilon_2 > 0$ be given. By the assumption, we have

$$\begin{aligned} \exists \delta > 0 \text{ s.t. } |x - a| < \delta &\Rightarrow |f(x)/g(x)| < \epsilon_1 \\ \exists N \in \mathbb{N} \text{ s.t. } n \geq N &\Rightarrow \mathbb{P}(|X_n - a| \geq \delta) \leq \epsilon_2. \end{aligned}$$

Thus,

$$\frac{f(x)}{g(x)} \geq \epsilon_1 \Rightarrow |x - a| \geq \delta$$

Substitute X_n for x and take probability on both side. Then,

$$\mathbb{P}\left(\frac{f(X_n)}{g(X_n)} \geq \epsilon_1\right) \leq \mathbb{P}(|X_n - a| \geq \delta) \leq \epsilon_2$$

□

3 Asymptotic Analysis in MLE